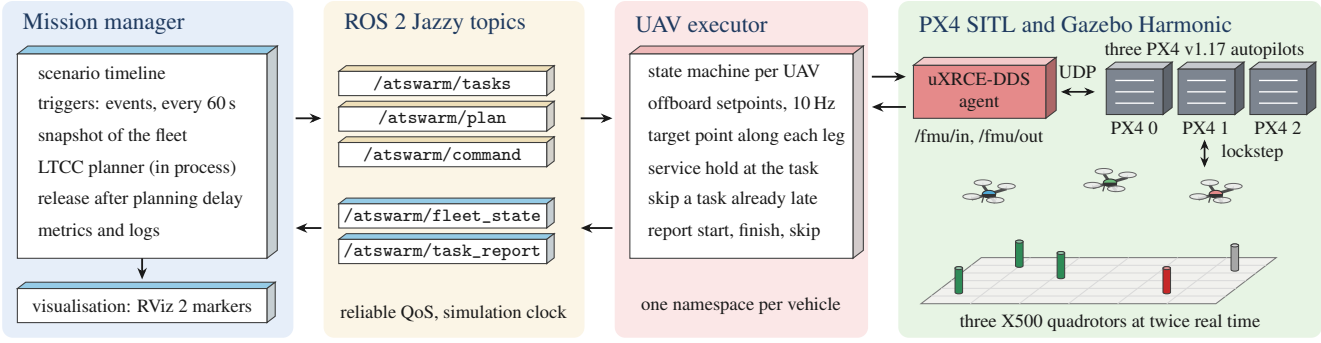

length scaled by 0.1
time scaled by 0.25
speed, energy, prices adjusted
utility of every flight compared with tier B on the same mission
Tier B mission: 10 km, 1800 s
same dimensionless ratios
Tier C mission: 1 km, 450 s, 3 UAVs
same seeds and disturbances


Mission manager
scenario timeline
triggers: events, every 60 s
snapshot of the fleet
LTCC planner (in process)
release after planning delay
metrics and logs
visualisation: RViz 2 markers
ROS 2 Jazzy topics
/atswarm/tasks
/atswarm/plan
/atswarm/command
/atswarm/fleet_state
/atswarm/task_report
reliable QoS, simulation clock
UAV executor
state machine per UAV
offboard setpoints, 10 Hz
target point along each leg
service hold at the task
skip a task already late
report start, finish, skip
one namespace per vehicle
PX4 SITL and Gazebo Harmonic
uXRCE-DDS agent
UDP
three PX4 v1.17 autopilots
/fmu/in, /fmu/out
PX4 0
PX4 1
PX4 2
lockstep
three X500 quadrotors at twice real time